

HONEYWELL SOFTWARE ENGINEERING TEAMS

AI Champions Programme

# Module 07

## MCP-Enabled Agentic Engineering Workflows and Reusable Engineering Skills

*Understanding MCP's architecture, transports, and JSON-RPC foundations, then connecting agents to approved enterprise systems under clear governance — and packaging what works into reusable skills.*

|                                                                                     |                            |                                                                                      |                               |                                                                                       |                                              |
|-------------------------------------------------------------------------------------|----------------------------|--------------------------------------------------------------------------------------|-------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------|
|  | <b>DURATION</b><br>2 Hours |  | <b>FORMAT</b><br>Hands-On Lab |  | <b>DELIVERED FOR</b><br>Honeywell Eng. Teams |
|-------------------------------------------------------------------------------------|----------------------------|--------------------------------------------------------------------------------------|-------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------|

# How We'll Spend These Two Hours

Six connected moves — from MCP's core architecture to a governed, reusable workflow.

01

MCP Architecture: Host, Client, Server

15 MIN

The foundational topology every MCP-enabled workflow is built on.

02

Transports & JSON-RPC: How Messages Actually Move

15 MIN

Stdio and Streamable HTTP transports, and the JSON-RPC 2.0 messages underneath both.

03

Key Concepts: Tools, Resources, Prompts & Sampling

15 MIN

The primitives a server exposes, and the one capability a server can ask of the client.

04

Connecting Agents to Enterprise Systems

15 MIN

Repositories, APIs, databases, documentation, and test infrastructure — through approved servers.

05

Governance: Permissions, Approvals & Failure Handling

15 MIN

What separates a governed MCP workflow from an ungoverned one.

06

Packaging Reusable Skills & Hands-On Lab

45 MIN

Build an MCP-enabled workflow end to end, then package it as a versioned, owned, reusable skill.

# Module 07 in the 12-Module Journey

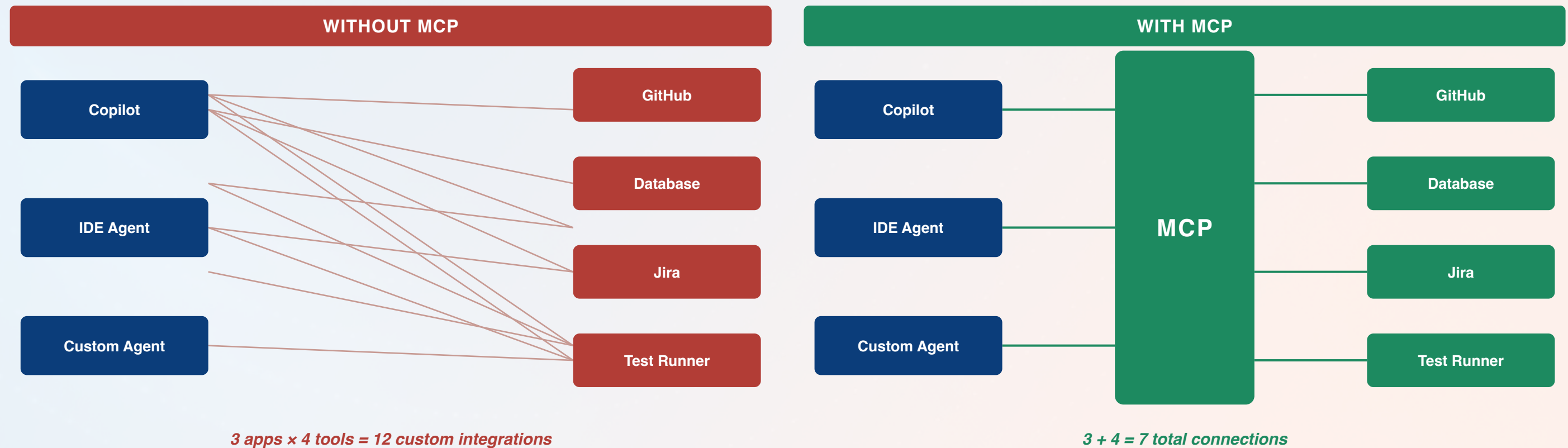
MCP is how everything built so far — Copilot, specs, tests — connects to real enterprise systems.



**Why this matters:** Module 08's PR automation and Module 09's studio workflows both assume agents already reach approved tools through governed MCP servers — this module is where that connection is built.

# Why MCP Exists: From $M \times N$ to $M + N$

*Without a shared protocol, every AI application needs custom code for every tool. MCP standardizes the connection.*

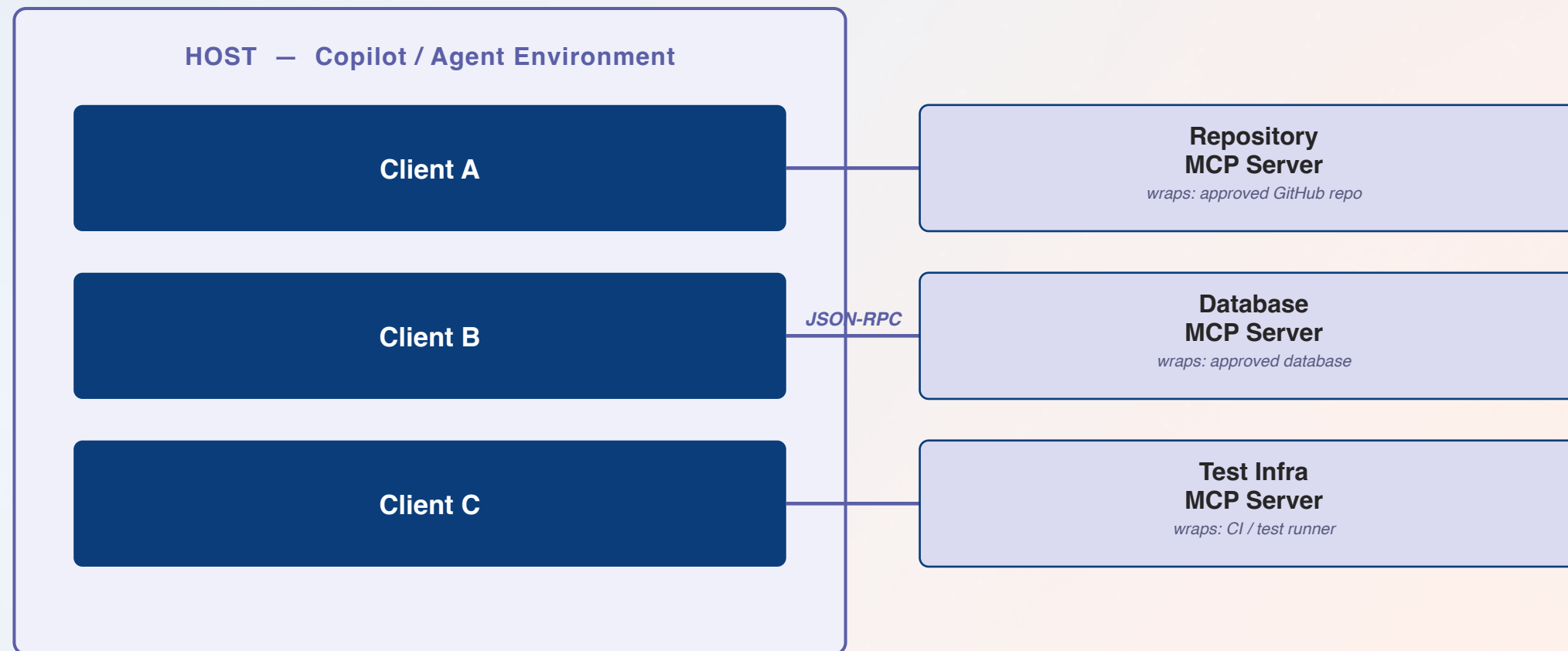


**Why this matters:** Every new tool your team wants to connect no longer means custom integration work for every AI application that needs it — build the MCP server once, and every MCP-compatible host can use it.



# MCP Architecture: Host, Client, Server

*One host, one or more clients, each holding a dedicated 1:1 connection to a server.*



**Why this matters:** Every server the agent uses — repository, database, test infrastructure — is reached through this same pattern: a dedicated client, a stable protocol, a clear boundary.

# Transports & JSON-RPC: How Messages Actually Move

*Two ways a client reaches a server — one message format underneath both.*

## STDIO TRANSPORT

- Server runs as a local subprocess
- Communicates over standard input/output
- No network exposure — process-to-process only
- Ideal for: local repo access, file system, CLI tools

## STREAMABLE HTTP TRANSPORT

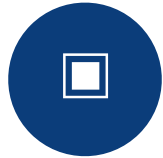
- Server runs remotely, reached over HTTP
- Supports streaming responses back to the client
- Requires authentication and network governance
- Ideal for: shared databases, internal APIs, services

EVERY MESSAGE, ON EITHER TRANSPORT, IS JSON-RPC 2.0

```
{
  "jsonrpc": "2.0",
  "id": 7,
  "method": "tools/call",
  "params": { "name": "run_tests", "arguments": { "suite": "checkout" } }
}
```

# Key Concepts: Tools, Resources, Prompts & Sampling

*Three things a server exposes, and one thing a server can ask of the client.*



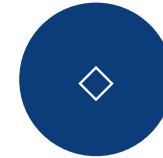
## Tools

Model-controlled functions the agent can invoke — run a test, query a database, open a PR. The AI decides when to call them.



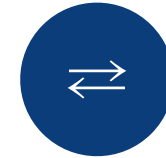
## Resources

Application-controlled data the host can read and attach as context — a file, a spec document, a log stream.



## Prompts

User-controlled templated interactions — reusable prompt structures a person selects, not the model.

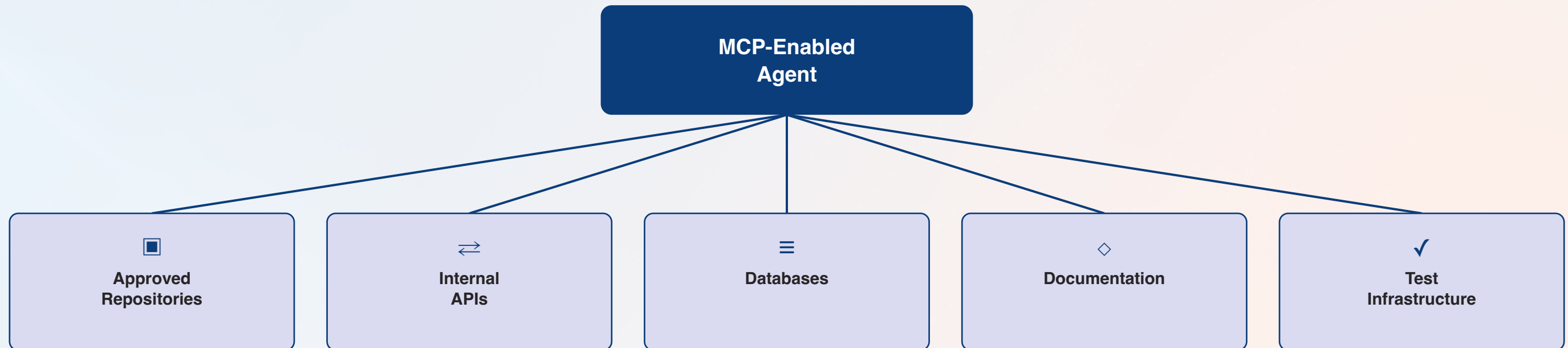


## Sampling

A server can ask the client's LLM to complete a step on its behalf — inverting the usual direction of control.

# Connecting Agents to Enterprise Systems

*Five categories of approved system an MCP-enabled agent can be connected to.*



**Why this matters:** Each connection is a separate, scoped MCP server — the agent's reach is exactly the union of the servers it's been given, nothing more.



# Governance: Permissions, Approvals & Failure Handling

*What separates a governed MCP workflow from an ungoverned one.*

| GOVERNED                                                                          | UNGOVERNED                                                                 |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <div><div></div>Tools scoped to approved servers, with explicit permissions</div> | <div><div></div>Any tool available to any agent, no permission model</div> |
| <div><div></div>Sensitive actions require human approval before executing</div>   | <div><div></div>No approval step before a tool call executes</div>         |
| <div><div></div>Failures are caught, logged, and surfaced clearly</div>           | <div><div></div>Failures fail silently or crash the whole workflow</div>   |
| <div><div></div>Every tool call is auditable — what ran, with what result</div>   | <div><div></div>No record of what a tool touched or changed</div>          |

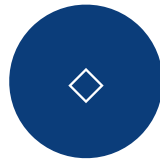
# Packaging Reusable Engineering Skills

*What worked once, turned into a versioned, owned, cross-project asset.*



## Specifications & Instructions

The spec templates and reusable instructions behind a workflow, packaged so the next project starts from them.



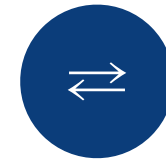
## Prompt Packs

Proven prompt structures for a recurring task, bundled and versioned rather than reinvented each time.



## Test / Review Heuristics

The checks a spec validator or PR reviewer applies, documented as a reusable, improvable heuristic set.



## Agent Behaviors

The tool sequence and decision logic an agent follows, packaged with clear ownership and a reuse model.

# Hands-On Lab Preview: Build a Workflow, Package a Skill

*The 45-minute activity that closes this module.*



**Pick a  
Workflow**

Repository analysis, spec validation, test execution, or PR evidence collection.



**Connect  
Via MCP**

Wire the agent to the approved server(s) the workflow needs.



**Run  
End-to-End**

Execute the workflow against real repository, test, or API data.



**Package the  
Skill**

Turn what worked into a reusable, versioned skill asset.



**Document  
Governance**

Record ownership, permissions, and the reuse model for the team.

**The throughline:** A workflow that only works once isn't a skill — packaging and governance are what make it reusable across projects.

## Module 07 Outcomes & What's Next

- ✓ The M × N integration problem MCP solves, and its host/client/server architecture, are both understood
- ✓ Stdio and Streamable HTTP transports, and JSON-RPC 2.0 underneath both, are clear
- ✓ Tools, Resources, Prompts, and Sampling are identified as the core MCP concepts
- ✓ Agents have been connected to approved repositories, APIs, databases, and test infrastructure
- ✓ Governance practices — permissions, approvals, failure handling — are understood
- ✓ An MCP-enabled workflow has been built and packaged as a reusable, governed skill



### NEXT MODULE

Module 08 — PR Process Automation, Quality Gates and LLM-as-Judge (1.5 hours). Configuring AI-assisted PR review against specification compliance, coding standards, and the test evidence built in Modules 06-07.



# Questions & Discussion

*Module 07 — MCP-Enabled Agentic Engineering Workflows and Reusable Engineering Skills*